



YEAR 11 YEARLY EXAMINATION

August 2006
Assessment 4

Mathematics

Outcomes referred to: P1, P2, P3, P4, P5, P6, P7, P8

General Instructions

- Working time – $2\frac{1}{2}$ hours
- 5 minutes reading time.
- Write using blue or black pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Begin a NEW question on a NEW page.

Total marks – 100

- Attempt questions 1-10
- Weighting of 40%
- Each question is worth 10 marks

Question 1:

(a) Express $\frac{17}{45}$ as a recurring decimal 1

(b) Evaluate correct to 2 decimal places $\sqrt{\frac{8.5-2.4}{3.2}} + (6.3)^3$. 2

(c) Show that 0.35 can be expressed as a common fraction by a method other than the use of a calculator. 2

(d) Simplify $\sqrt{147}$. 1

(e) Express $6 \times \sqrt{\frac{1}{3}}$ as a complete surd. 1

(f) Rationalise the denominator of $\frac{1}{\sqrt{5}+2}$. 2

(g) Simplify $-45x^3y^2 \div 9x^3y^5$. 1

Question 2:

- (a) Simplify $3m^2 - 3mn - m + n$. 1
- (b) Expand and simplify $a(a+1) - 3(2a+1)$. 1
- (c) Expand and simplify $(5x-14)(2x+5)$ 1
- (d) Factorise $36c^2 - 49d^2$ 1
- (e) Factorise $\frac{a^3}{8} - 1$ 1
- (f) Factorise $x^2 - 21x - 72$ 1
- (g) (i) Factorise $6x^2 - 29x + 28$ 2
- (ii) Solve $6x^2 - 29x + 28 = 0$ 2

Question 3:

(a) Simplify $\frac{x^2 - 7x + 6}{x^2 - 36}$ **2**

(b) Express the following as an integer $5^3 \times 2^4 \times 10^{-3}$ **1**

(c) Solve for m : $(2m - 5)^2 = 16$ **2**

(d) Graph on the number line the solution of: $\frac{3x}{5} - \frac{2x}{3} > -2$ **3**

(e) Simplify $\frac{3a}{2b} - \frac{5a}{b}$ **2**

Question 4:

(a) Solve $(x+1)^2 = 4x$

1

(b) Solve: $\frac{5x-3}{2} = x+2$

2

(c) Solve simultaneously $\begin{matrix} x+5y=16 \\ 2x-y=7 \end{matrix}$ by the use of the *elimination method*

3

(d) Solve $|2x+5| < 3$

3

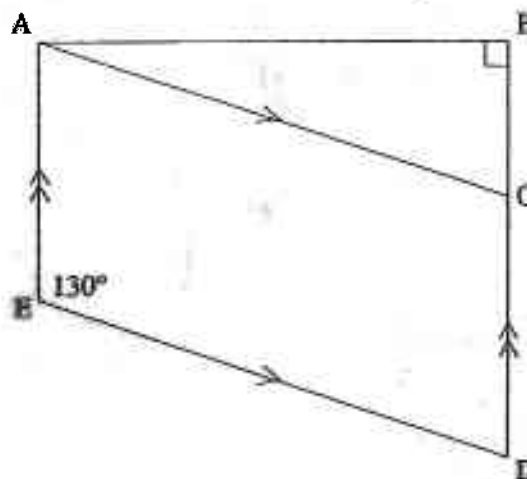
(e) Solve $22 \leq 5x-3 \leq 32$

1

Question 5:

- (a) In the diagram $AE \parallel BD$, $AC \parallel ED$, $\angle AED = 130^\circ$ and $\angle ABC = 90^\circ$.

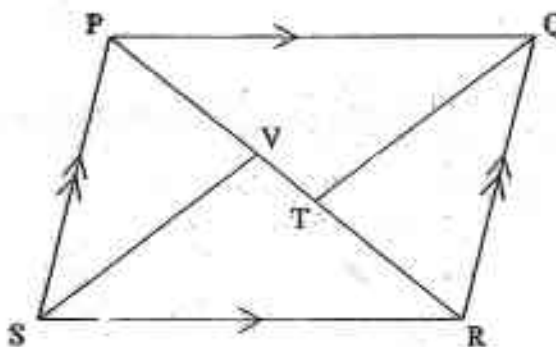
NOT TO SCALE



- (i) Copy this diagram onto your answer sheet.
 (ii) Find the size of $\angle BAC$ giving reasons.

2

(b)



NOT TO SCALE

PQRS is a parallelogram. TQ bisects $\angle PQR$ and VS bisects $\angle PSR$.

- (i) Copy this diagram onto your answer booklet.
 (ii) State why $\angle PQR = \angle PSR$.
 (iii) Prove that $\triangle PVS$ and $\triangle RTQ$ are congruent.
 (iv) Hence find the length of TV if $PR = 20\text{cm}$ and $TR = 8\text{cm}$.

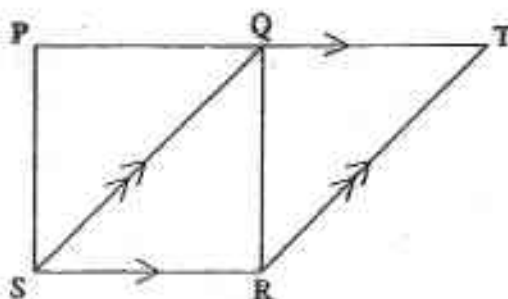
1

3

2

- (c) PQRS is a square with $PQ = 1\text{unit}$. Find the perimeter of PTRS.

3



NOT TO SCALE

Question 6:

- (a) In the diagram the line L_1 has equation $3x - 2y + 3 = 0$ and the line L_2 passes through the point $P(0, 8)$ and $Q(12, 0)$.

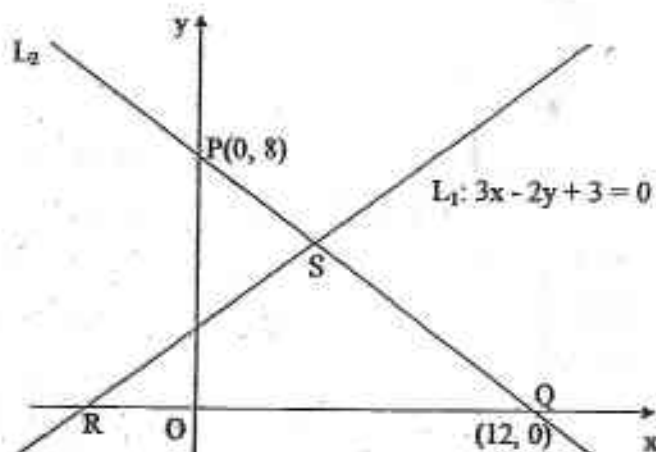


FIGURE NOT TO SCALE

- | | | |
|-------|---|---|
| (i) | L_1 cuts the x-axis at R. Find the co-ordinates of R. | 1 |
| (ii) | Find the gradient of the line L_1 . | 1 |
| (iii) | Show the line L_1 is perpendicular to the line L_2 . | 1 |
| (iv) | Show the equation of the line L_2 is: $2x + 3y - 24 = 0$. | 2 |
| (v) | Find the coordinates of S, the point of intersection of the lines L_1 and L_2 . | 1 |
| (vi) | Hence calculate the area of $\triangle QRS$. | 2 |

- (b) What is the domain and Range of $y = -\sqrt{9 - x^2}$ 2

Question 7:

(a) Sketch the curve $y = |2x| - 4$, showing all essential features.

3

(b) Given:

$$f(x) = \begin{cases} x+1, & x \geq 1 \\ -1, & -1 < x < 1 \\ 1-x, & x \leq -2 \end{cases}$$

(i) Evaluate $f(1)$

1

(ii) Evaluate $f\left(\frac{1}{2}\right)$

1

(c) Shade the common region(s) in the XY-plane indicated by the inequalities $x + 2y \geq 8$ and $y < 7$.

3

(d) For $90^\circ < A < 180^\circ$ find A if $\sin A = \frac{\sqrt{3}}{2}$

2

Question 8:

- (a) Find the exact value of: $\cot(-60^\circ)\operatorname{cosec}(240^\circ)$. 2
- (b) AB and CD are two vertical buildings with their bases A and C on horizontal ground. The height of AB is 30m. The angle of elevation of B as seen from C is 25° and the angle of elevation of D as seen from A is 40° .
- (i) the horizontal distance between the buildings 2
- (ii) the height of CD. 2
- (iii) The angle of depression of B as seen from D. 1
- (c) A ship sailing in a direction $065^\circ T$ observes two lighthouses in a line due North. After travelling 4km, one of the lighthouses bears $285^\circ T$ and the other $315^\circ T$. Calculate the distance between the lighthouses. 3

Question 9:

(a) (i) Find from first principles, the derivative of $f(x) = x^2 - 2$. **2**

(ii) Hence or otherwise, find $f'(x)$ at $x = -2$. **1**

(b) Find the derivative of the following equations.

(i) $y = 2x^3 + 3x$ **1**

(ii) $f(x) = \frac{2x^5 - x + 1}{x^3}$ **2**

(iii) $(x-3)\sqrt{x-3}$ **2**

(iv) $y = \frac{2x+7}{2x-3}$ **2**

Question 10:

- (a) Find the equations of the tangent and normal to the parabola $y = 2x^2 - 5x + 1$ at the point on the parabola where the gradient is 3. 3

- (b) Find p if the roots of the equation $3x^2 + px - 12 = 0$ differ by 4. 2

- (c) Express $x^2 + 2x - 2$ in the form $ax(x+1) + bx^2 + c(x+1)$. 3

- (d) Find the equation of the family of parabolas having:

- (i) the line $x = 2$ as axis and vertex $(2, 1)$. 1

- (ii) The line $x = -2$ as axis and focal length $\frac{1}{2}$ unit. 1

Question 1:

a) $0.3\dot{7}$ ✓

b)

c) $x = 0.3555\ldots$

$$10x = 3.5555$$

$$9x = 3.2$$

$$90x = 32$$

$$x = \frac{32}{90}$$

$$x = \frac{16}{45}$$

d) $7\sqrt{3}$ ✓

e) $\sqrt{36 \times \frac{1}{3}}$
 $= \sqrt{12}$ ✓

f) $\frac{1}{\sqrt{5} + 2} \times \frac{\sqrt{5} - 2}{\sqrt{5} - 2}$
 $= \frac{\sqrt{5} - 2}{5 - 4}$ ✓
 $= \sqrt{5} - 2$ ✓

Question 2:

(a) $3m(m-n) - (m-n)$
 $= (3m-1)(m-n)$ ✓

(b) $a^2 + a - 6a - 3$
 $= a^2 - 5a - 3$ ✓

(c) $10x^2 - 28x + 25x - 70$
 $= 10x^2 - 3x - 70$ ✓

(d) $(6c-7d)(6c+7d)$ ✓

(e) $\left(\frac{a}{2} - 1\right)\left(\frac{a^2}{4} + \frac{a}{2} + 1\right)$ ✓

(f) $(x+3)(x-24)$ ✓

g) (i) $(3x-4)(2x-7)$ ✓

(ii) $x = \frac{4}{3}$ ✓
 $x = \frac{7}{2}$ ✓

Question 3:

a) Simplify

$$\frac{(x-1)(x/6)}{(x/6)(x+6)}$$

$$= \frac{x-1}{x+6}$$

b) $5^3 \times 2^4 \times (5 \times 2)^{-3}$

$$= 5^3 \times 2^4 \times 5^{-3} \times 2^{-3}$$

$$= 5^0 \times 2^1$$

$$= 2$$

c) $2m-5 = \pm 4$

$$2m = 5 \pm 4$$

$$m = \frac{5}{2} \pm \frac{4}{2}$$

$$m = \frac{5}{2} \pm 2$$

d) $\frac{9x-10x}{15} > -2$

$$-x > -30$$

$$x < 30$$

30

(e) $\frac{3a-10a}{2b}$

$$= \frac{-7a}{2b}$$

Question 4:

(a) $x^2 + 2x + 1 = 4x$

$$x^2 - 2x + 1 = 0$$

$$(x-1)(x-1) = 0$$

$$x = 1$$

(b) $5x - 3 = 2x + 4$

$$3x = 7$$

$$x = \frac{7}{3}$$

(c) $2x + 10y = 32$ — (1)

$$2x - y = 7$$
 — (2)

$$11y = 25$$

$$y = \frac{25}{11}$$

$$2x - \frac{25}{11} = 7$$

$$2x = 9\frac{3}{11}$$

$$x = 4\frac{7}{11}$$

$$(d) \quad 2x + 5 < 3$$

$$2x < -2$$

$$\boxed{x < -1}$$

$$-2x - 5 < 3$$

$$-2x < 8$$

$$\boxed{x > -4}$$

$$2) \quad 25 \leq 5x \leq 35$$

$$5 \leq x \leq 7$$

$$x \leq 7$$

$$x \geq 5$$

Question 5:

$$a)(ii) \quad \angle EDC = 50^\circ \text{ — co-interior } \angle\text{'s in } \parallel \text{ lines}$$

$$\angle ACB = 50^\circ \text{ — corresponding } \angle\text{'s in } \parallel \text{ lines}$$

$$\therefore \text{ In } \triangle ABC = 90 + 50 + 90$$

$$180^\circ = 90 + 140$$

$$\angle BAC = 40^\circ$$

(b) (ii) opposite $\angle$'s in a parallelogram are equal.

$$\therefore \angle PQR = \angle PSR$$

$$(iii) \quad \angle SPR = \angle QRT$$

alt angles in $\parallel$ lines

$PS = QR$ — equal sides in a parallelogram

$$\angle PSV = \angle RQT$$

opposite $\angle$'s equal that have been bisected.

$\therefore (AAS)$

$$\triangle PVS \equiv \triangle RTQ$$

(iv) $\therefore$ since

$$PR = 20 \text{ cm}$$

$$\text{and } PV = RT$$

congruent triangles.

$$\therefore TV = 20 - 16 = 4 \text{ cm.}$$

$$(c) \quad PS = PQ = SR = 1 \text{ unit.}$$

$$RT = \sqrt{1^2 + 1^2}$$

$$RT = \sqrt{2}$$

$\therefore QT = SR$ — opposite sides of a parallelogram are equal.

$$\therefore P = 4 + \sqrt{2} \text{ units}$$

Question 6:

a) $3x - 2y + 3 = 0$

x-axis (let $y = 0$)

$$3x = -3$$

$$x = -1$$

$$\therefore R(-1, 0) \checkmark$$

ii) Rearranging.

$$2y = 3x + 3$$

$$y = \frac{3}{2}x + \frac{3}{2}$$

$$m = \frac{3}{2} \checkmark$$

iii) $L_2 \text{ m} = -\frac{8}{12} = -\frac{2}{3}$

$$\therefore -\frac{2}{3} \times \frac{3}{2} = -1 \checkmark$$

$$\therefore L_2 \neq L_1$$

i) $y - 0 = -\frac{2}{3}(x - 12)$

$$y = -\frac{2}{3}x + \frac{24}{3}$$

$$y = -\frac{2}{3}x + 8 \checkmark$$

$$3y = -2x + 24$$

$$2x + 3y - 24 = 0 \checkmark$$

(v) $3x - 2y = -3$ — (1)

$$2x + 3y = 24$$
 — (2)

$$6x - 4y = -6$$
 — (1)

$$6x + 9y = 72$$
 — (2)

$$-13y = -78$$

$$\boxed{y = 6}$$

$$3x - 12 = -3$$

$$3x = 9$$

$$\boxed{x = 3}$$

$$S(3, 6) \checkmark$$

(vi) $A = \frac{1}{2}bh$ $\checkmark$

$$A = \frac{1}{2} \times 13 \times 6$$

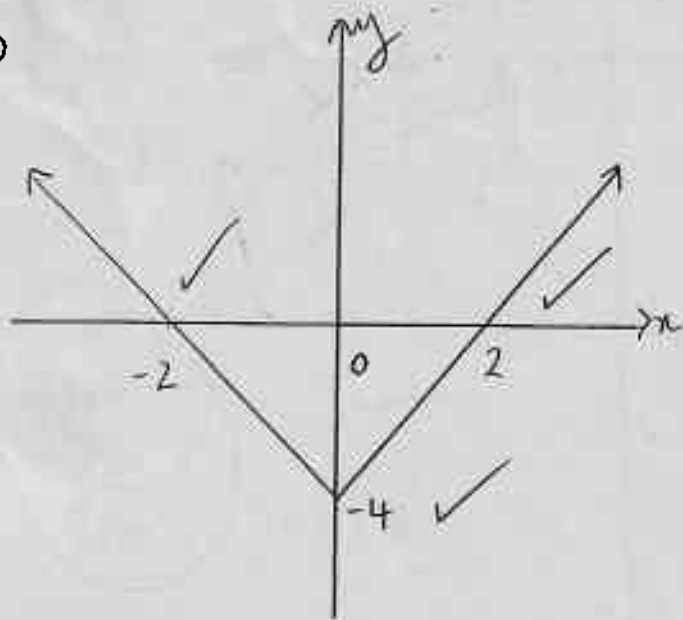
$$\boxed{A = 39 \text{ units}^2} \checkmark$$

b) Domain: $-3 \leq x \leq 3$ $\checkmark$

$$\text{Range: } -3 \leq y \leq 0 \checkmark$$

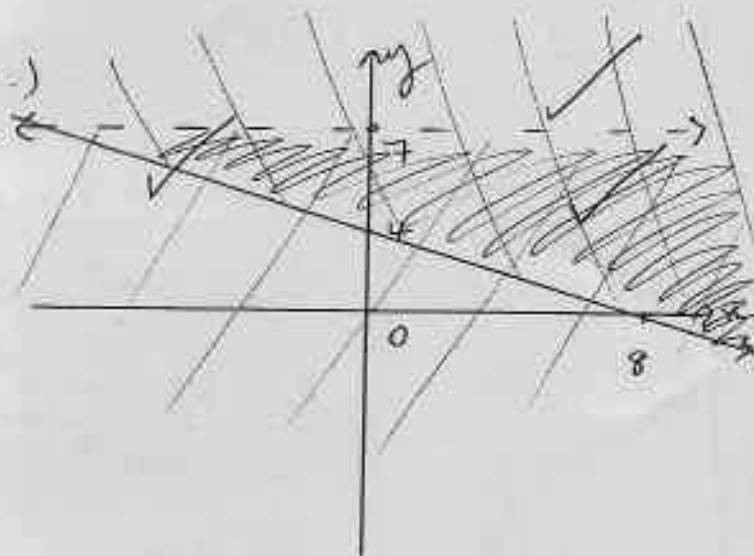
Question 7:

(a)



c) (i) $f(1) = 2$ ✓

(ii) $f(\frac{1}{2}) = -1$ ✓



d)

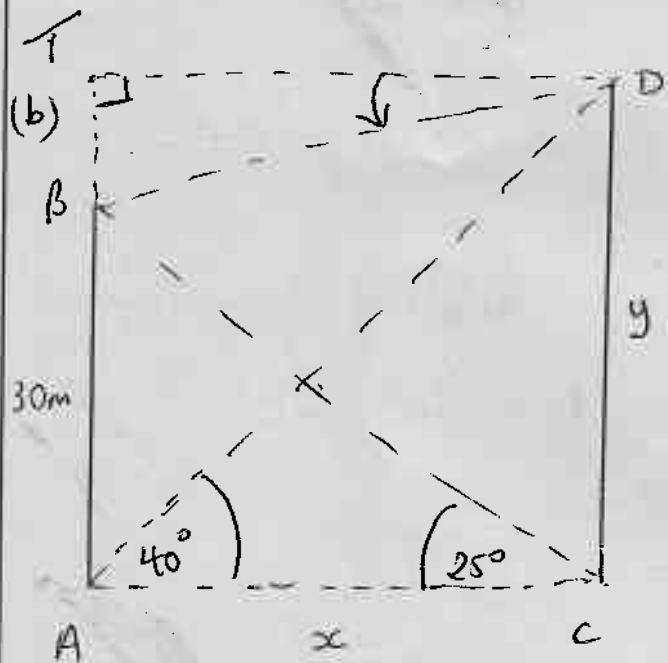


$\sin 60^\circ = \frac{\sqrt{3}}{2}$ ✓

$A = 120^\circ$ ✓

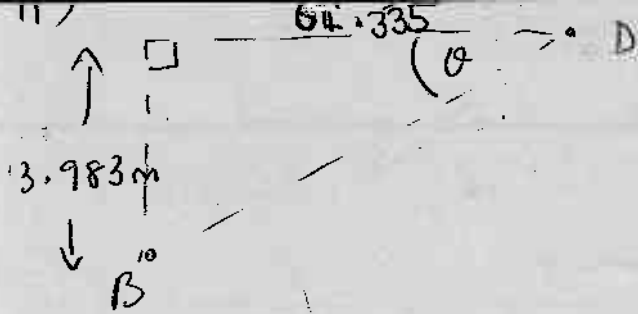
Question 8:

$$\begin{aligned} (a) & \frac{1}{\tan(-60^\circ)} \times \frac{1}{\sin(240^\circ)} \\ &= \frac{1}{-\sqrt{3}} \times \frac{1}{-\sqrt{3}/2} \checkmark \\ &= -\frac{1}{\sqrt{3}} \times -\frac{2}{\sqrt{3}} \\ &= \frac{2}{3} \checkmark \end{aligned}$$



$$\begin{aligned} (i) \quad \frac{30}{x} &= \tan 25^\circ \\ x &= \frac{30}{\tan 25^\circ} \\ x &= 64.335 \text{ m} \end{aligned}$$

$$\begin{aligned} (ii) \quad \frac{y}{64.335} &= \tan 40^\circ \\ y &= 64.335 \times \tan 40^\circ \\ y &= 53.983 \text{ m} \end{aligned}$$



$$\tan \theta = \frac{23.983}{64.335}$$

$$\theta = 20^\circ 27'$$

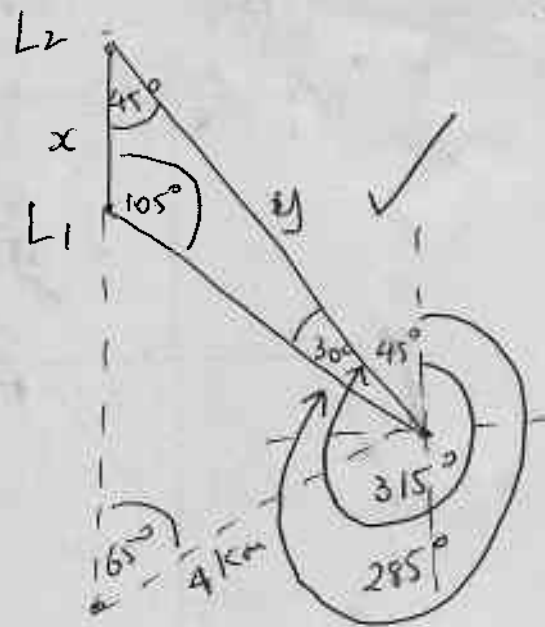
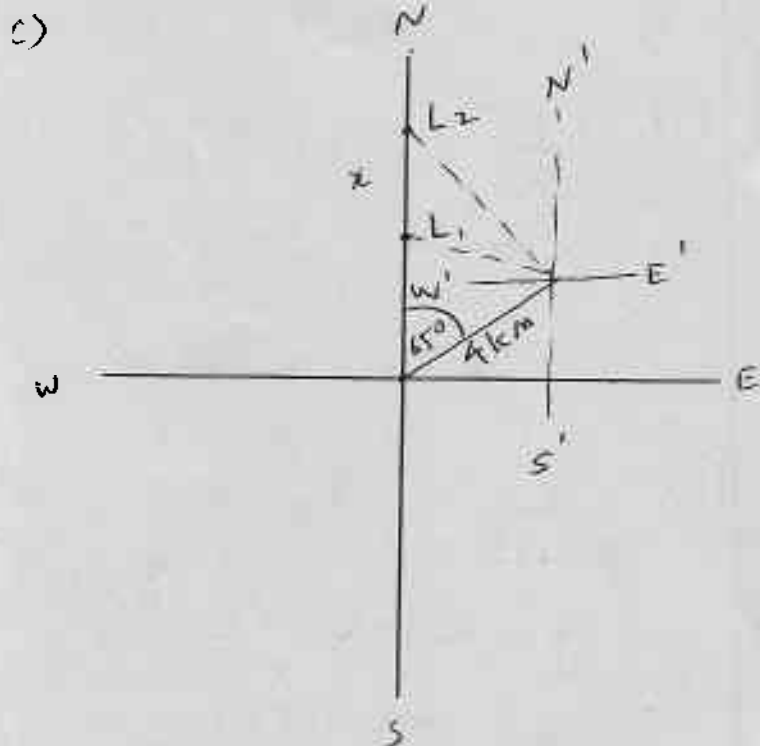
$$\theta = 20^\circ$$

$\angle ADT = 40^\circ$ — alt $\angle$'s
in || lines

Angle of depression

$$= 40^\circ - 20^\circ$$

$$\theta = 20^\circ$$



Part 1 : $\frac{y}{\sin 65^\circ} = \frac{4}{\sin 45^\circ}$

$$y = \frac{4 \sin 65^\circ}{\sin 45^\circ}$$

$$y = 5.1269$$

$$\frac{x}{\sin 30^\circ} = \frac{5.1269}{\sin 105^\circ}$$

$$x = \frac{5.1269 \times \sin 30^\circ}{\sin 105^\circ}$$

$$x = 2.654$$

Question 9:

$$2) (i) \quad f(x+h) = x^2 + 2xh + h^2 - 2$$

$$f(x) = x^2 - 2$$

$$m = \frac{f(x+h) - f(x)}{h}$$

$$m = \frac{(x^2 + 2xh + h^2 - 2) - (x^2 - 2)}{h}$$

$$m = \frac{2xh + h^2}{h}$$

$$m = 2x + h$$

$$\therefore m = 2x + h$$

$$\lim_{h \rightarrow 0}$$

$$m = 2x$$

$$i) \therefore \text{at } x = -2$$

$$m = -4$$

$$3) (i) \quad \frac{dy}{dx} = 6x^2 + 3$$

$$(ii) \quad f(x) = 2x^2 - \frac{1}{x^2} + \frac{1}{x^3}$$

$$f'(x) = 4x + 2x^{-3} - 3x^{-4}$$

$$f'(x) = 4x + \frac{2}{x^3} - \frac{3}{x^4}$$

$$(iii) \quad \frac{d}{dx} = (x-3) \times \frac{1}{2} (x-3)^{-1/2} + \sqrt{x-3} \times 1$$

$$\frac{d}{dx} = \frac{x-3}{2\sqrt{x-3}} + \sqrt{x-3}$$

$$(iv) \quad \frac{dy}{dx} = \frac{(2x-3) \times 2 - (2x+7) \times 2}{(2x-3)^2}$$

$$\frac{dy}{dx} = \frac{4x-6-4x-14}{(2x-3)^2}$$

$$\frac{dy}{dx} = \frac{-20}{(2x-3)^2}$$

Question 10:

$$(a) \frac{dy}{dx} = 4x - 5$$

$$3 = 4x - 5$$

$$8 = 4x$$

$$\boxed{2 = x}$$

$$\text{at } (2, -1) \\ m = 3$$

tangent:

$$y + 1 = 2(x - 2)$$

$$\boxed{y = 2x - 5}$$

Normal:

$$y + 1 = -\frac{1}{3}(x - 2)$$

$$y = -\frac{1}{3}x - \frac{1}{3}$$

$$b) \alpha - \beta = 4$$

$$\boxed{\alpha = \beta + 4}$$

$$\alpha + \beta = -\frac{P}{3}$$

$$\beta + 4 + \beta = -\frac{P}{3}$$

$$2\beta + 4 = -\frac{P}{3}$$

$$2\beta = -\frac{P}{3} - 4$$

$$\boxed{\beta = -\frac{P}{6} - 2}$$

$$\alpha\beta = -4$$

$$\beta(\beta + 4) = -4$$

$$\beta^2 + 4\beta + 4 = 0$$

$$(\beta + 2)(\beta + 2) = 0$$

$$\beta = -2$$

$$-2 = -\frac{P}{6} - 2$$

$$0 = -P$$

$$0 = P$$

(c)

$$x^2 + 2x - 2 = ax(x+1) + bx^2 + c(x+1)$$

$$\text{RHS } ax^2 + ax + bx^2 + cx + c$$

$$(a+b)x^2 + (a+c)x + c$$

$$\therefore a + b = 1$$

$$a + c = 2$$

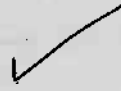
$$\boxed{c = -2}$$

$$\boxed{a = 4}$$

$$\boxed{b = -3}$$

$$4x(x+1) - 3x^2 - 2(x+1)$$

$$(d) (i) (x-2) = \pm 4a(y-1)$$



$$(ii) (x+2)^2 = \pm 2(y-y_1)$$

